

Metformin

Product Information

BRAND / TRADE NAMES

Glucophage | Glumetza | Fortamet | Riomet

HOW SUPPLIED

Metformin HCl 500 mg tablets: white to off-white, round, film-coated tablets. Available in bottles of 100 tablets and bottles of 500 tablets.

STORAGE & HANDLING

Store at 20-25 degrees C (68-77 degrees F). Excursions permitted to 15-30 degrees C (59-86 degrees F). Keep container tightly closed. Protect from light and moisture.

KEY DRUG FACTS

Generic Name: metformin hydrochloride

Drug Class: Biguanide Antidiabetic Agent

Manufacturer: Bristol-Myers Squibb Company

Route: Oral

Rx Status: Prescription Only

QR CODE

For healthcare professionals only. Refer to full prescribing information before clinical use. Data sourced from US FDA OpenFDA.

Route Oral

Rx Only

Form Tablet

metformin hydrochloride
Metformin

Biguanide Antidiabetic Agent

Metformin

metformin hydrochloride

Biguanide Antidiabetic Agent

Route

Oral

Class

Biguanide Antidiabetic

Rx Only

Yes

Mfd by Bristol-Myers Squibb Company

PRESCRIBING INFORMATION BROCHURE

Clinical Information

INDICATIONS & USAGE

Metformin HCl tablets are indicated as an adjunct to diet and exercise to improve glycemic control in adults and pediatric patients 10 years of age and older with type 2 diabetes mellitus.

CONTRAINDICATIONS

Contraindicated in patients with: Severe renal impairment (eGFR below 30 mL/min/1.73 m²). Hypersensitivity to metformin. Acute or chronic metabolic acidosis, including diabetic ketoacidosis with or without coma.

CLINICAL USE NOTES

- Monitor renal function before and during therapy.
- Assess eGFR at initiation and periodically thereafter.
- Counsel patients on signs of lactic acidosis.
- Vitamin B12 monitoring recommended annually.
- Avoid use in conditions predisposing to hypoxia.

PATIENT POPULATION

Adults: 18 years and older
Pediatric: 10–16 years
Geriatric: Use with caution
Pregnancy: Consult physician

Drug Overview

DESCRIPTION

Metformin hydrochloride is an oral antihyperglycemic drug used in the management of type 2 diabetes mellitus. It is not chemically or pharmacologically related to any other class of oral antihyperglycemic agents. Metformin decreases hepatic glucose production, decreases intestinal absorption of glucose, and improves insulin sensitivity.

DOSAGE & ADMINISTRATION

Adults: Starting dose 500 mg orally twice daily or 850 mg once daily with meals. Increase in increments of 500 mg weekly or 850 mg every 2 weeks, up to a maximum dose of 2550 mg per day in divided doses. Doses above 2000 mg may be better tolerated given 3 times daily with meals.
Pediatrics (10-16 years): Starting dose 500 mg twice daily with meals; maximum 2000 mg per day.

DOSAGE QUICK REFERENCE

Adults: 500 mg twice daily with meals
Max Dose: 2550 mg per day
Pediatric: 500 mg twice daily (10-16 yrs)
Max Peds: 2000 mg per day
Titration: Increase by 500 mg weekly

MECHANISM OF ACTION

- Decreases hepatic glucose production (gluconeogenesis).
- Reduces intestinal glucose absorption.
- Improves peripheral insulin sensitivity.
- Does not stimulate insulin secretion — low hypoglycemia risk.

Safety Information

WARNINGS & PRECAUTIONS

Lactic Acidosis: Postmarketing cases have resulted in death, hypothermia, hypotension, and resistant bradyarrhythmias. Symptoms include malaise, myalgias, respiratory distress, somnolence, and abdominal pain. Discontinue if lactic acidosis is suspected. Vitamin B12 Deficiency: Metformin may lower vitamin B12 levels; measure hematological parameters annually.

ADVERSE REACTIONS

Most common adverse reactions (greater than 5%): diarrhea, nausea and vomiting, flatulence, asthenia, indigestion, abdominal discomfort, and headache. Decreased Vitamin B12 levels observed in approximately 7% of patients in controlled clinical trials.

MONITORING PARAMETERS

- Renal function (eGFR) — before & during therapy
- Vitamin B12 — every 2–3 years
- HbA1c — every 3–6 months
- Fasting blood glucose — regularly
- Signs of lactic acidosis — ongoing

NOTABLE DRUG INTERACTIONS

Iodinated contrast: Hold 48 hrs before/after
Alcohol: Increases lactic acidosis risk
Insulin/sulfonylurea: Monitor for hypoglycemia
Topiramate: Increased metformin levels